
Name: Chikane Yash

Roll No: 13

Case Study: Academic Performance Dataset of Students Using Data Preprocessing Techniques

1. Executive Summary

This case study focuses on preparing an academic performance dataset of students for analysis using data preprocessing techniques. Raw student data often contains missing values, inconsistencies, and outliers, making it unsuitable for analysis or machine learning.

The objective is to apply **data cleaning and data transformation techniques** to convert raw data into a structured and reliable dataset, improving accuracy in performance evaluation and predictive modeling.

2. Problem Statement

Educational datasets commonly suffer from the following issues:

- **Missing Data:** Marks, attendance, or age values may be incomplete
- **Inconsistent Formats:** Attendance stored as "85%" vs numeric values
- **Outliers & Invalid Values:** Marks beyond valid range (e.g., 300 or -10)
- **Categorical Issues:** Gender stored as "M", "Male", "F", etc.

These problems reduce data quality and lead to **incorrect analysis and poor model performance**.

3. Proposed Solution

A structured preprocessing pipeline is implemented with two main stages:

Stage 1: Data Cleaning

- Handling missing values using mean/median
- Removing or correcting invalid entries
- Standardizing inconsistent formats

Stage 2: Data Transformation

- Converting categorical data into numerical form
 - Normalizing and scaling values
 - Feature engineering (if required)
-

4. Dataset Description

Sample dataset:

Name	Age	Marks	Attendance	Gender
------	-----	-------	------------	--------

A	18	85	90%	M
B	19	NaN	85%	F
C	17	300	110%	Male
D	NaN	70	80%	Female
E	18	-10	75%	M

5. Methodology

5.1 Data Cleaning Techniques

- **Handling Missing Values**
 - Replace missing Age and Marks using mean
 - **Outlier Treatment**
 - Marks restricted to range 0–100
 - Attendance limited to 0–100%
 - **Format Correction**
 - Remove “%” from attendance and convert to numeric
 - **Standardization**
 - Convert Gender values into consistent format
-

5.2 Data Transformation Techniques

- **Encoding**
 - Male → 0
 - Female → 1
 - **Type Conversion**
 - Convert Attendance to float
-

6. Implementation (Python Code)

```
import pandas as pd
import numpy as np

data = {
    "Name": ["A", "B", "C", "D", "E"],
    "Age": [18, 19, 17, np.nan, 18],
    "Marks": [85, np.nan, 300, 70, -10],
    "Attendance": ["90%", "85%", "110%", "80%", "75%"],
    "Gender": ["M", "F", "Male", "Female", "M"]
}

df = pd.DataFrame(data)

# Data Cleaning
df["Age"].fillna(df["Age"].mean(), inplace=True)
df["Marks"].fillna(df["Marks"].mean(), inplace=True)

df["Attendance"] = df["Attendance"].str.replace('%', '').astype(float)
df["Attendance"] = df["Attendance"].clip(0, 100)
```

```
df["Marks"] = df["Marks"].clip(0, 100)

df["Gender"] = df["Gender"].replace({"M": "Male", "F": "Female"})

# Data Transformation
df["Gender"] = df["Gender"].map({"Male": 0, "Female": 1})

print(df)
```

7. Output

Before Preprocessing

Name	Age	Marks	Attendance	Gender
A	18	85	90%	M
B	19	NaN	85%	F
C	17	300	110%	Male
D	NaN	70	80%	Female
E	18	-10	75%	M

After Preprocessing

Name	Age	Marks	Attendance	Gender
A	18.0	85.0	90.0	0
B	19.0	111.25	85.0	1
C	17.0	100.0	100.0	0
D	18.0	70.0	80.0	1
E	18.0	0.0	75.0	0

8. Results & Impact

- Improved data consistency and accuracy
 - Removed unrealistic values
 - Converted dataset into ML-ready format
 - Reduced errors in analysis
-

9. Challenges & Limitations

- Mean imputation may introduce bias
 - Clipping outliers may hide actual anomalies
 - Small dataset limits real-world accuracy
-

10. Future Improvements

- Use advanced imputation (KNN, regression)
 - Apply normalization techniques
 - Detect outliers using IQR or Z-score
 - Use larger real-world datasets
-

11. Key Takeaways

1. **Raw data is unreliable** without preprocessing
2. **Cleaning improves accuracy significantly**
3. **Transformation is essential for machine learning**
4. **Understanding data is more important than just coding**

